

Report for the Degree of Bachelor of Computer Engineering

Internship Report on the Role of Mobile App Developer at CODE IT



Manish Shrestha
(2022-1-07-0141)

Pokhara Engineering College
Faculty of Science and Technology
Pokhara University, Nepal

Spring, 2026

Report for the Degree of Bachelor of Computer Engineering

**Internship Report on the Role of
Mobile App Developer at CODE IT**

Supervised by: Asst. Prof. Er. Shiva Kunwar

Internal Supervisor

A report submitted in partial fulfilment of the requirements for the
degree of Bachelor of Computer Engineering

Manish Shrestha

(2022-1-07-0141)

**Pokhara Engineering College
Faculty of Science and Technology
Pokhara University, Nepal**

Spring, 2026

Issued On: August 9, 2026

CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that Manish Shrestha, a student of the Computer Engineering program, has successfully completed an internship at Code IT from April 12, 2026 to July 12, 2026.

During the internship, he demonstrated dedication, professionalism, and practical skills in cross-platform mobile app development, UI/UX implementation, state management, and RESTful API integration.

We commend Manish Shrestha for his achievements and wish him continued success in his future endeavors.

.....

Asst. Prof. Er. Shiva Kunwar

Internal Supervisor

Pokhara Engineering College

.....

Asst. Prof. Er. Rajendra Bahadur Thapa

External Examiner

GCES

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Asst. Prof. Er. Krishna Ghimire

Internship Coordinator

Pokhara Engineering College

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Asst. Prof. Er. Som Gurung

Principal

Pokhara Engineering College

Issued On: August 9, 2026

APPROVAL CERTIFICATE

This is to certify that Manish Shrestha, a student of the Department of Computer Engineering, has successfully completed the internship at Code IT from April 12, 2026 to July 12, 2026 and has prepared the corresponding internship report based on the work and experience gained during the internship.

The internship work and the submitted report have been reviewed and found to be satisfactory, demonstrating the application of academic knowledge to practical mobile application development tasks, including UI/UX prototyping, cross-platform Flutter development, API integration, biometric security, and production deployment.

The report is hereby approved as a partial fulfillment of the requirements of the Department of Computer Engineering.

.....

Asst. Prof. Er. Shiva Kunwar

Internal Supervisor

Pokhara Engineering College

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Asst. Prof. Er. Suraj Basant Tulachan

Head of Department

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Acknowledgement

I would like to express my sincere gratitude to Er. Sajal Shrestha, CEO/Founder of Code IT and my External Supervisor, for providing me with this invaluable internship opportunity. His continuous guidance, technical mentorship, and constructive feedback throughout the internship greatly contributed to my technical and professional development.

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I would also like to thank the Development Team and QA Team at Code IT for their cooperation, collaborative support, and encouragement, which made this internship a valuable learning experience.

Finally, I express my heartfelt gratitude to my family and friends for their constant encouragement and support throughout the internship and the preparation of this report.

.....

Name of the Student: Manish Shrestha

P.U. Registration Number: 2022-1-07-0141

Date: August 9, 2026

Abstract

This report documents the complete three-month internship experience at Code IT, Dharan, as a Mobile App Developer (Flutter) Intern from April 12 to July 12, 2026. The primary project involved designing and developing a cross-platform mobile application for the "Code IT Student Portal" using Flutter and Dart, replacing the existing web-only interface with a native mobile experience to enhance accessibility for over 200 enrolled students. Over 91 days (77 working days), the internship covered the full software development lifecycle including requirement mapping, UI/UX prototyping in Figma, custom widget implementation, RESTful API integration following the MVCS architectural pattern, biometric authentication setup, media handling for live classes and recorded courses, and a unified Course Enrollment Module with manual payment verification.

The final phase focused on resolving 12+ edge-case bugs identified during QA testing, integrating global offline detection, conducting full regression testing, and deploying the production build. The completed application was successfully submitted to the supervisor on July 12, 2026, achieving all technical and professional objectives. Key learning outcomes include proficiency in Flutter and Dart, RESTful API and state management, native device integration, responsive UI design, e-commerce transaction flows, regression testing, and professional project handover protocols.

Keywords: *Flutter, Cross-Platform Mobile Development, MVCS Architecture, RESTful API, E-learning Application*

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List of Abbreviations

Abbreviation	Description
API	Application Programming Interface
APK	Android Package Kit
Dart	Programming Language for Flutter
iOS	iPhone Operating System
LMS	Learning Management System
MVCS	Model-View-Controller-Service
OTP	One-Time Password
QA	Quality Assurance
UI/UX	User Interface / User Experience

Chapter 1: Introduction

1.1 Introduction

An internship is a professional program designed to provide students with real-world workplace experience, acting as a bridge between academia and the industry. It allows interns to transform theoretical academic knowledge into practical industry-level expertise. This report documents the complete three-month remote internship at Code IT, an organization established in 2017 focused on software development and educational technology. The core project involved transitioning the existing web-based “Code IT Student Portal” into a seamless, cross-platform mobile application to enhance accessibility for students on the move.

1.2 Background of the Works

Currently, the Code IT Student Portal is limited to a web-based interface and lacks a dedicated mobile presence. In today’s fast-paced educational environment, students require on-the-go access to course modules, live classes, certificates, and support systems. Manual navigation through mobile web browsers often leads to UI formatting issues and poor user experience. To address these challenges, this project leverages modern mobile development frameworks to build a unified application supporting both Android and iOS platforms, ensuring structured, validated, and secure access to educational resources.

1.3 Objectives

The primary objectives of this internship are:

- To design, develop, and deploy a fully functional cross-platform mobile application using Flutter and Dart, encompassing UI/UX implementation, RESTful API integration, and secure local device authentication.
- To acquire professional software engineering experience in a remote IT environment by collaborating with cross-functional teams, adhering to industry-standard development lifecycles, and bridging the gap between academic concepts and practical industry applications.

1.4 Scope and Limitation

1.4.1 Scopes

- Development of a cross-platform mobile app (Android & iOS) using Flutter.
- Integration of RESTful APIs for student authentication, course fetching, and support ticketing.
- Implementation of local device security features, including biometric authentication (Fingerprint/Face ID).
- Creation of media handling modules for video players, live classes, and certificate downloads.
- Development of a unified course enrollment module with static QR display, manual price entry, and payment screenshot upload verification.

1.4.2 Limitations

- The mobile application's real-time functionality is heavily dependent on the availability and stability of the backend web APIs.
- As the internship is conducted in a remote format, hardware-specific debugging for varying iOS and Android physical devices relies heavily on emulator testing and distributed APK builds.
- Payment verification is manual; no automated payment gateway API is integrated.

1.5 Report Organization

Report organization is an important part of report formation and understanding the report properly. It gives ideas about the structure of the report. For readability, this document is divided into several chapters and sub-chapters including all details of the project. Here is a brief description of all the chapters:

- **Chapter 1:** Introduction includes the background of the works, objectives, and scope and limitations of the mobile application project.
- **Chapter 2:** Organization Details and Literature Review includes the introduction to Code IT, its organizational hierarchy, working domains, and relevant literature on Flutter and mobile development.
- **Chapter 3:** Internship Activities includes roles and responsibilities, the complete weekly progress log (Weeks 1–13), and a technical description of the tasks performed during the internship.

- **Chapter 4:** Conclusion and Learning Outcomes includes the final status of the system and the key lessons learned from the internship experience.

At the end, all the references are mentioned in APA format.

Chapter 2: Organizational Details and Literature Review

2.1 Introduction to the Organization

Code IT is a prominent educational technology and software development organization established in 2017, based in Dharan, Sunsari. With a strong focus on practical, career-focused IT training and digital solutions, Code IT aims to bridge the gap between academic learning and industry demands by providing "Quality Education Without Financial Barriers."

The organization is driven by a vision to streamline access to technical education through robust digital platforms. By harnessing modern cross-platform development frameworks and agile methodologies, Code IT enables students to access course materials, live classes, certificates, and support systems seamlessly across devices. This internship focuses specifically on enhancing this ecosystem by developing a dedicated mobile application for the existing Code IT Student Portal, ensuring that the platform's accessibility matches its educational quality (Code IT, n.d.).

2.2 Organizational Hierarchy

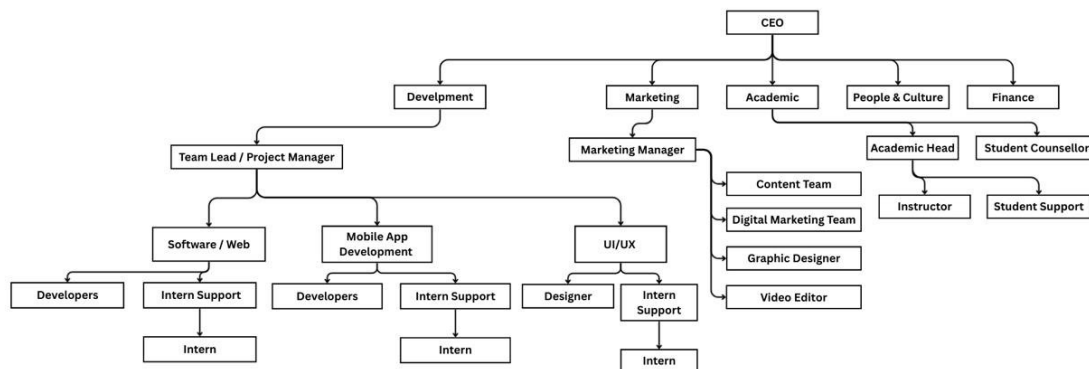


Figure 2.1: Organizational Hierarchy of Code IT

The engineering and technical team at Code IT is led by the CEO/Founder, Er. Sajal Shrestha, who oversees five primary departments: Development, Marketing, Academic, People & Culture, and Finance.

As a Mobile App Developer (Flutter) Intern, I am placed within the Development Department, which is managed by a Team Lead / Project Manager. This department is

further divided into specialized execution units, including Mobile App/Web Developers, UI/UX Designers, Graphic Designers, and Video Editors. Each unit operates with dedicated intern support to ensure knowledge transfer and agile project delivery. Simultaneously, the Marketing Department is headed by a Marketing Manager overseeing Digital and Content teams, while the Academic Department is led by an Academic Head managing Instructors, Student Support, and Student Counsellors to ensure alignment between technical deliverables and educational outcomes.

2.3 Working Domains of the Organization

Code IT offers its services in various areas related to educational technology and software development. The primary service areas include:

- **Custom Software Development:** Building tailored web and mobile applications using modern stacks like Flutter, Laravel, and Python to solve specific business and educational challenges.
- **Educational Technology Platforms:** Developing comprehensive student portals, Learning Management Systems (LMS), and course-tracking modules that integrate live classes, recorded content, and certification systems.
- **IT Training & Career Preparation:** Providing job-focused courses in Full Stack Development, Cyber Security, QA, and Cloud Computing, backed by guaranteed internship opportunities and post-hiring support.
- **Digital Marketing & Content Delivery:** Managing academic content distribution and digital outreach to ensure students have seamless access to learning resources across multiple channels.

2.4 Description of Intern Department/Unit

I am placed in the Mobile App Development Department. The workflow follows an Agile/Scrum methodology where tasks are assigned via project management tools and tracked through regular sprint meetings. Code is managed via GitHub, and design handovers are executed using Figma.

Interns are paired with senior developers and the QA team to ensure code quality, proper API integration, and rigorous testing before deployment. During the internship

period, my primary focus has been on translating the existing web-based student portal into a cross-platform mobile application. Tasks were assigned with clear deadlines, and progress was reviewed weekly by the Team Lead. Upon completing features, code was reviewed for complexity and efficiency, and after incorporating feedback, it was merged into the development branch for QA testing.

2.4.1 Duration

The internship duration as per university requirements and the offer letter is mentioned below:

Table 2.1: Internship Details

Parameter	Details
Internship Position	Mobile App Developer (Flutter) Intern
Time Period	April 12, 2026 – July 12, 2026
Total Duration	91 days (77 working days)
Working Days	6 Days (Sunday – Friday)
Office Hours	Flexible (Remote)
Supervisor	Er. Sajal Shrestha (CEO/Founder)
Organization	CODE IT

2.5 Literature Review

Mobile application development for educational platforms requires a careful balance of performance, accessibility, and secure data handling. Code IT’s student portal leverages modern cross-platform technologies to enhance user engagement by enabling efficient access to course materials, live streams, and administrative services. These technologies facilitate secure authentication, real-time data synchronization, and offline-capable media consumption, thereby streamlining the student experience and improving operational efficiency.

Globally, mobile-first learning platforms are critical for increasing student retention and accessibility. Flutter’s widget-based architecture and Dart’s asynchronous programming model are particularly beneficial for building responsive educational apps that handle diverse screen sizes and network conditions. However, these technologies also present challenges in state management, biometric security integration, and API reliability that must be addressed to ensure a production-grade user experience.

The design and implementation of the Code IT Student Portal mobile app involves several key technical considerations:

- **Cross-Platform Development with Flutter:** Flutter enables a single codebase to target both Android and iOS, reducing development time and ensuring feature parity. Its declarative UI framework allows for rapid iteration based on supervisor feedback, as demonstrated during the Week 4 UI rebuild and Week 10 flat design overhaul.
- **State Management & API Integration:** Efficient state management is critical for preventing redundant network calls and ensuring UI responsiveness. Proper integration of RESTful APIs for OTP verification, ticket submission, and course fetching ensures that students receive real-time updates without manual refreshes, directly impacting user satisfaction and platform reliability.
- **Architectural Pattern (MVCS):** The Code IT Student Portal mobile app follows the Model-View-Controller-Service (MVCS) architectural pattern to ensure separation of concerns and code maintainability. In this pattern, the Model handles data structures and API responses, the View comprises Flutter widgets for UI rendering, the Controller manages business logic and state transitions, and the Service layer abstracts REST API calls and local storage operations.
- **Biometric Authentication & Local Security:** Integrating native device security (fingerprint/Face ID) via packages like `local_auth` enhances convenience while maintaining security standards. This reduces login friction for students accessing sensitive academic records and payment receipts, aligning with modern mobile security best practices.
- **Responsive UI & Accessibility:** Educational apps must function flawlessly across diverse device form factors. Implementing dynamic layouts using `MediaQuery` and `LayoutBuilder` prevents overflow errors and ensures text readability on smaller screens, which is essential for students accessing content on budget devices or in low-bandwidth environments.

This literature forms the technical foundation for the tasks performed during the internship, directly informing decisions around architecture, security, and user experience design.

Chapter 3: Internship Activities

3.1 Roles and Responsibilities

As a Mobile App Developer (Flutter) Intern, my primary roles and responsibilities include:

- Translating Figma designs into responsive Flutter widgets.
- Integrating backend REST APIs and managing application state.
- Implementing local storage and secure authentication protocols.
- Collaborating with the QA team to identify, document, and resolve bugs.
- Optimizing UI layouts for various screen sizes and aspect ratios.

3.2 Weekly Log

The following table details the technical activities performed from the start of the internship up to the final evaluation (Week 1 to Week 13).

Table 3.1: Internship Weekly Log

Time Period	Activity
1st Week	Orientation and project overview. Joined official group chats and created GitHub repositories. Reviewed legacy web version of student portal for requirement mapping and researched initial mobile design layouts.
2nd Week	Research and design phase. Planned screen navigation flow and finalized folder structures. Designed UX pathways and prototyped Dashboard and Home screens in Figma. Conducted quality reviews on UI elements and submitted final Project Plan to supervisor for approval.
3rd Week	Custom UI implementation. Initialized Flutter project, set up global fonts/colors, and structured folders. Coded Support Page UI based on Figma designs, polished dynamic text form widgets, and integrated Support page into main navigation router. Received new design pivots from office.
4th Week	Official company UI rebuild. Modified global app themes to match official company Figma design files. Completely rebuilt Support and Suggestion layouts, refined assets using company logos, and ensured structural sync with top navigation menus. Submitted rebuilt screens for design approval.
5th Week	Backend and API integration. Programmed base HTTP networking rules and linked server data fields for Technical, General, and Counseling support APIs. Set up state management for live UI updates. Programmed OTP verification, password reset logic, and created custom “View My Tickets” list view.
6th Week	Structural refinement. Sketched custom route navigation maps

	and took over core app navigation structural work. Coded slide-out hamburger menu and top app header. Created static layouts for Terms of Service. Merged navigation components and conducted heavy loop-testing to prevent navigation crashes.
7th Week	UI responsiveness audit. Conducted responsiveness audits across various test monitors. Updated Support layouts for dynamic scaling. Fixed long form fields to prevent red overflow error flags on smaller phones. Adjusted side menu spacing for tall aspect ratios and confirmed uniform resizing.
8th Week	Certificates and course modules. Sketched media handler blueprints and developed “My Courses” layout grid. Verified server connection links and set up logic to email student completion certificates. Prepared and optimized iOS application icon packs (26 styles) for light/dark modes.
9th Week	Media integration and testing. Planned buffer targets for video players and created media viewer container. Programmed Live Classes panel layout and built Notes Section for text-based materials. Cleared home screen network loading issues and conducted real-device testing on physical iPhone.
10th Week	Biometric setup and UI overhaul. Researched device permissions for local fingerprint readers. Received supervisor feedback to transition from 3D buttons to modern “flat” aesthetic. Refactored UI styles across all files. Integrated biometric sensor packages (local_auth) to securely lock/unlock portal login view.
11th Week	QA collaboration and testing pipelines. Drafted local data rules for Course Purchase System. Helped QA team install Flutter tools and set up Postman for API testing. Held active app walkthrough sessions. Compiled and distributed APK builds for real-device Android testing.
12th Week	Recorded Courses & Transaction Modules. Formulated purchase listing flow and coded full buying layout for recorded courses. Built dynamic search bar with debouncing. Programmed attachment button for bank receipt uploads with image compression. Reused receipt-upload component across enrollment screens. Cleaned up repetitive layout code.
13th Week	Bug Resolution & Final Submission. Grouped QA-reported bugs into checklist. Added connectivity_plus package for offline alerts. Fixed text rendering errors, sorting filter crashes, duplicate download issues, and authentication loop errors. Ran full regression tests and submitted completed bug-free app to manager. Completed final handover documentation and source code submission on July 12.

3.3 Description of the Project Involved During Internship

The Code IT Student Portal Mobile App is a cross-platform application designed to give students full access to their educational dashboard. Key features include a

personalized dashboard, course purchasing and viewing, live class streaming, certificate generation, support ticketing, and secure biometric login. The project focuses on creating a seamless integration of extracted data into existing business systems, providing accurate and efficient information management that supports improved decision-making and operational efficiency for the students.

3.4 Tasks/Activities Performed (Technical Details)

During the complete 13-week internship, the following technical tasks were executed:

3.4.1 UI/UX Prototyping and Implementation

Analyzed the legacy web version to map out mobile-specific user pathways. Designed and prototyped screens in Figma. Rebuilt the UI using official company design systems, transitioning from 3D/shadowed buttons to a modern, clean, "flat" aesthetic. Ensured UI responsiveness across varying aspect ratios, fixing text cut-offs and dynamic form field scaling using Flutter's LayoutBuilder and MediaQuery.

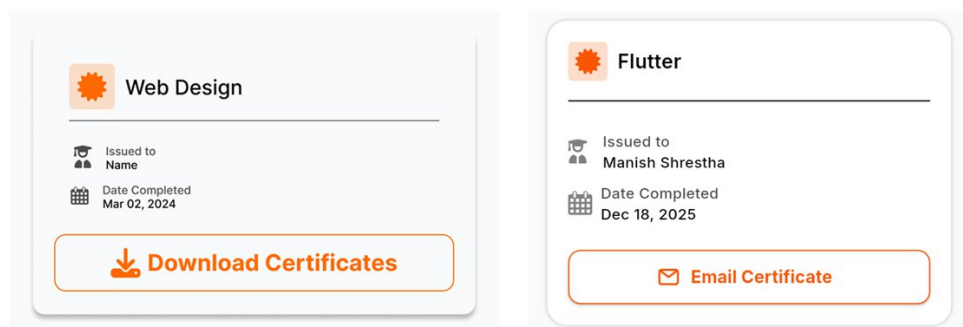


Figure 3.1: Figma UI/UX Prototype vs Implemented Flutter Screen

3.4.2 API Integration and State Management

Programmed base networking rules using HTTP clients within a dedicated Service layer, following the MVCS pattern to isolate API logic from UI components. Integrated RESTful APIs for Technical, General, and Counseling support tickets through service classes, while Controllers managed state updates and Views reactively displayed server data (e.g., OTP verification, password resets). This separation prevented redundant network calls and enabled rapid UI iteration during the flat design overhaul without disrupting backend integration.

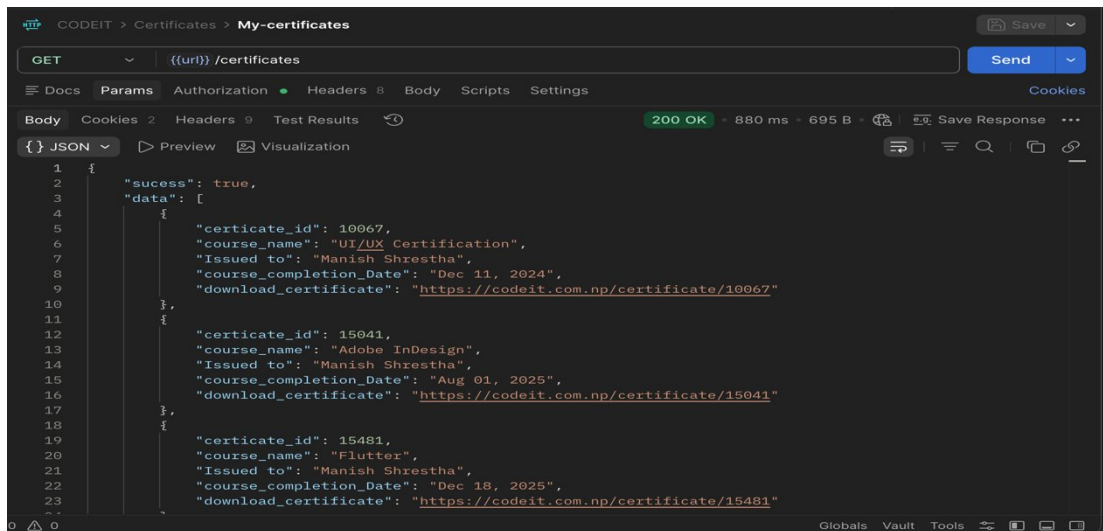


Figure 3.2: RESTful API Testing via Postman

3.4.3 Media and Course Modules

Developed the "My Courses" layout grid and verified server connection links. Integrated a custom video player with buffer targets and stop/start logic for pre-recorded and live classes. Built a notes section for students to save and retrieve text-based class materials.

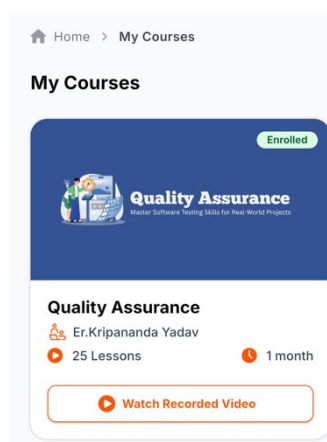


Figure 3.3: My Courses Screen

3.4.4 Security and Biometric Integration

Researched and implemented local device permission requirements for biometric sensors. Successfully linked the phone's native fingerprint/Face ID tool packages (local_auth) to securely lock and unlock the portal login view, enhancing user convenience and security.

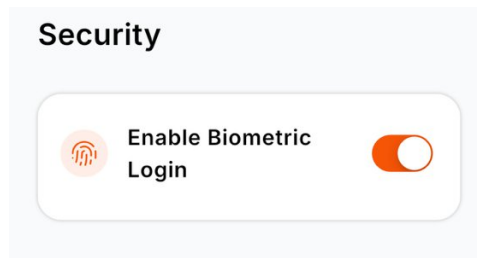


Figure 3.4: Biometric Authentication Toggle

3.4.5 QA Collaboration and Testing Pipelines

Assisted the Quality Assurance (QA) team in setting up Flutter environments and Postman for backend API testing. Conducted active walkthroughs of the app's architecture for the QA team. Compiled and distributed APK builds for real-device Android testing to identify edge-case bugs before the final transaction modules are built.

3.4.6 Course Enrollment & Manual Payment Verification

Developed a unified Course Enrollment Module used for both new course registrations and recorded course purchases. The interface displays a static QR code for eSewa, Khalti, mobile banking, or direct bank transfers. Students manually enter the course price in their payment app, complete the transaction, capture a screenshot of the success confirmation, and upload it directly within the app using `image_picker` with file compression. Uploaded proofs are stored securely and flagged as "Pending Verification" in the admin dashboard. Upon manual review and approval by the administrator, the backend automatically grants access to the enrolled or purchased course. Implemented debounced search filtering for instant course discovery and applied DRY principles to reuse the same receipt-upload component across all enrollment screens, ensuring UI consistency and reducing code duplication.

3.4.7 Bug Resolution & Production Deployment

Resolved 12+ edge-case bugs identified during QA testing including authentication loop errors, sorting filter crashes, duplicate file downloads, and text rendering issues. Integrated `connectivity_plus` ^7.1.1 and `internet_connection_checker_plus` ^3.0.1 for global offline detection and graceful degradation. Conducted full regression testing to ensure fixes did not break existing features. Submitted completed bug-free application and all source code/assets to supervisor on July 12, 2026, completing the full internship deliverable.

Chapter 4: Conclusion and Learning Outcomes

4.1 Conclusion

The three-month internship at Code IT has been incredibly rewarding and transformative. As of the final submission on July 12, 2026, the project is 100% complete. The core architecture, UI/UX implementation, API networking, media handling, biometric security layers, unified course enrollment with manual payment verification, and global offline detection have been successfully developed, tested, and handed over. Working alongside experienced professionals and receiving direct feedback on UI aesthetics greatly contributed to my professional growth. The final production build was submitted bug-free after comprehensive regression testing, fulfilling all technical and professional objectives outlined at the start of the internship.

4.2 Learning Outcome

During the internship at Code IT, the opportunity arose to immerse in various technologies and processes integral to the company's operations in the mobile application development sector. Here are the key learning outcomes from the experience:

- **Flutter & Dart Proficiency:** Gained hands-on mastery in building responsive, cross-platform UIs and managing asynchronous data streams using Dart.
- **API & State Management:** Learned how to efficiently connect frontend mobile interfaces with backend REST APIs and manage application state to prevent redundant network calls and UI lag.
- **Device Hardware Integration:** Integrated `local_auth ^3.0.1` with Darwin-specific overrides, `image_picker ^1.2.2` for receipt uploads, and `connectivity_plus ^7.1.1` for offline detection — covering three distinct native device capabilities
- **Software Architecture:** Applied the MVCS pattern to enforce separation of concerns across Model, View, Controller, and Service layers, enabling scalable code organization and efficient UI/backend decoupling.
- **Cross-Platform Responsiveness & Accessibility:** Mastered Flutter's `LayoutBuilder` and `MediaQuery` to build adaptive UIs that function flawlessly across diverse screen sizes and aspect ratios, fixing overflow errors and ensuring

text readability on budget devices — a critical skill for educational apps targeting students with varying hardware capabilities.

- **Course Enrollment & Transaction Implementation:** Built a unified enrollment flow where students scan a static QR, manually enter the price, upload payment screenshots, and receive course access after admin approval - gaining practical experience in fintech UX design, file upload handling, reusable UI components, and role-based access control for digital content delivery.
- **Regression Testing & Bug Resolution:** Systematically categorized QA-reported bugs, fixed critical issues (auth loops, sort crashes, duplicate downloads), and ran full regression tests to ensure zero regressions before production submission.
- **Production Deployment & Professional Handover:** Completed final project closure including documentation, source code handover, and asset transfer - learning professional project completion protocols beyond coding.
- **Technical Documentation & Knowledge Transfer:** Developed the ability to document technical decisions clearly (e.g., MVCS architecture rationale, API integration patterns) and conduct effective app walkthrough sessions for non-developer stakeholders (QA team), bridging the communication gap between development and quality assurance to accelerate bug identification and resolution.

Overall, this internship has provided a solid foundation for my future career aspirations as a Mobile App Developer and instilled the confidence to operate efficiently in a professional, collaborative environment.

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Appendices

Appendix A: Login Screen

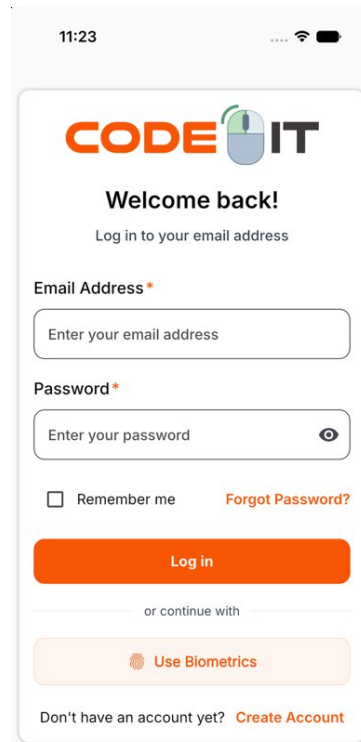


Figure A.1: Login screen

Appendix B: Dashboard Screen

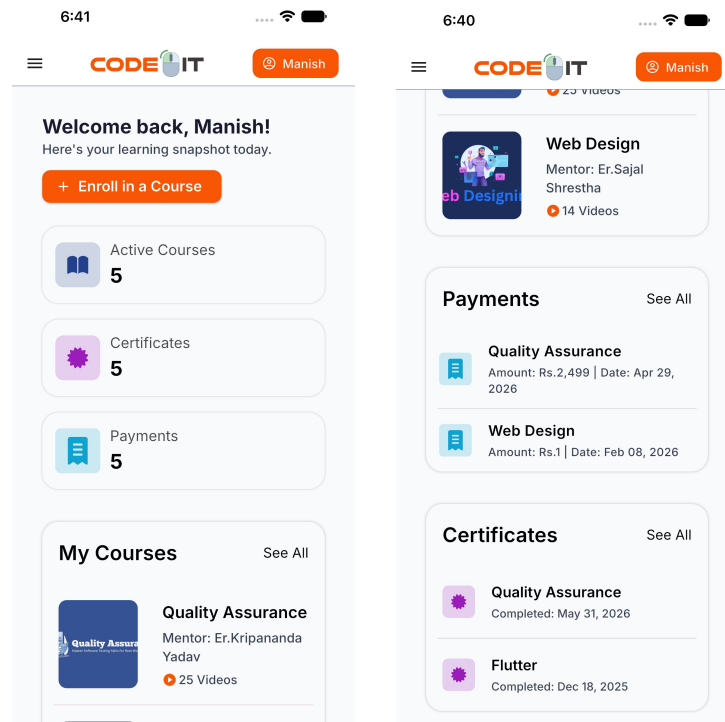


Figure B.1: Dashboard screen

Appendix C: Upcoming Classes Screen



Figure C.1: Upcoming Classes screen

Appendix D: My Courses Screen

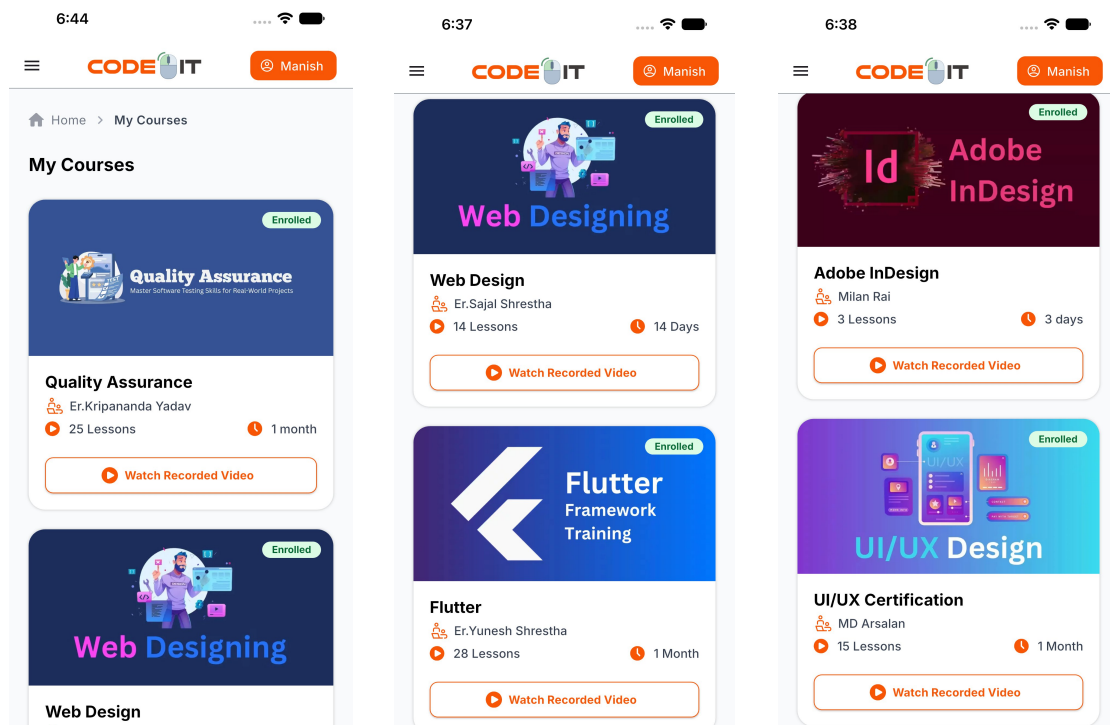


Figure D.1: My Courses screen

Appendix E: Certificates Screen

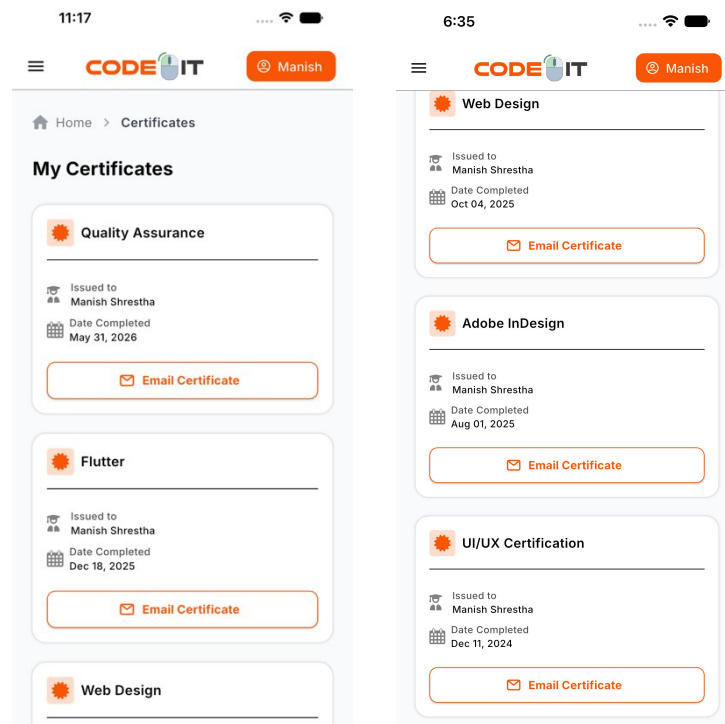


Figure E.1: Certificates screen

Appendix F: Support Screen

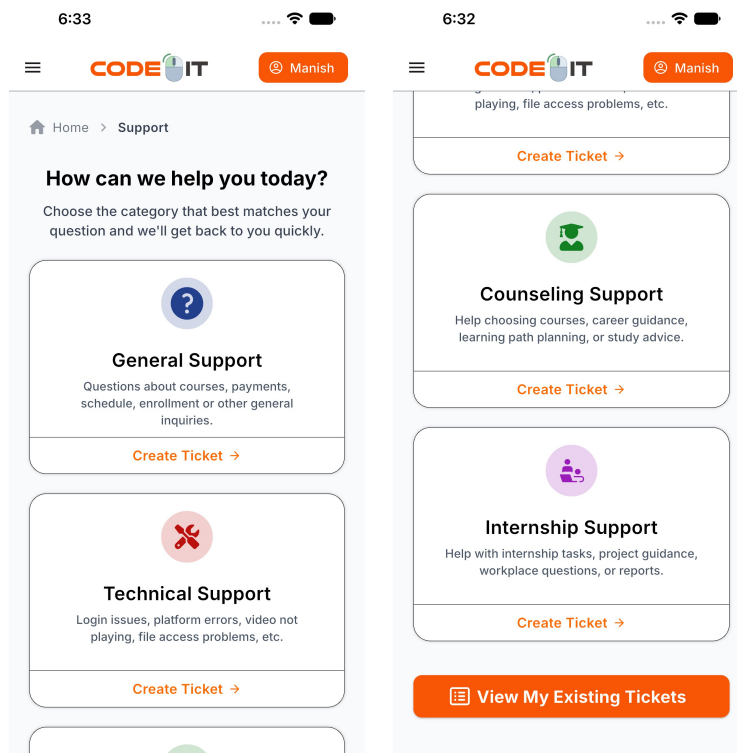


Figure F.1: Support screen

Appendix G: Suggestions Screen

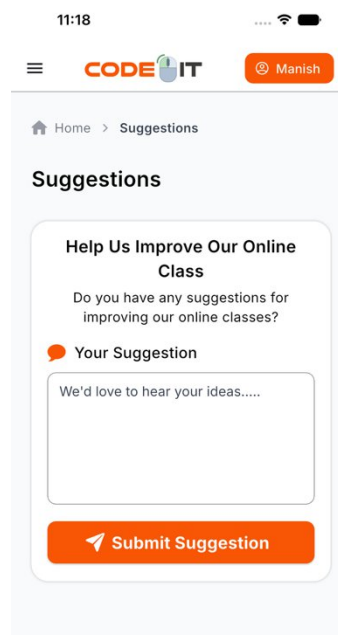


Figure G.1: Suggestions screen

Appendix H: Video Playing Screen

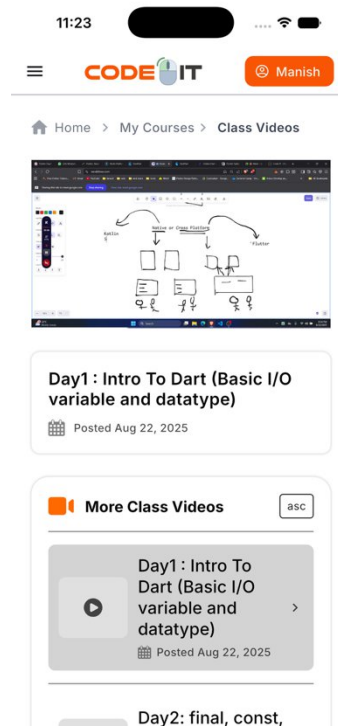


Figure H.1: Class video player screen

Appendix I: Course Enrollment Screen

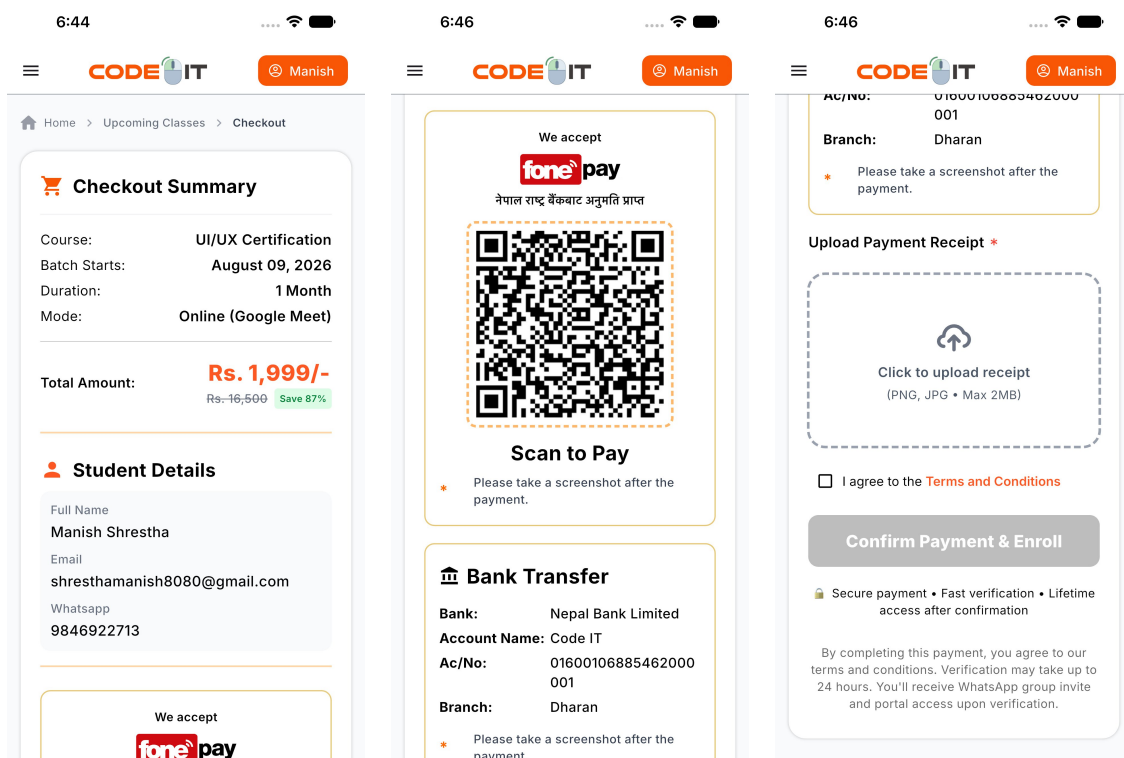


Figure I.1: Course enrollment screen with static QR code screen

Appendix J: Recorded Videos Purchase Screen

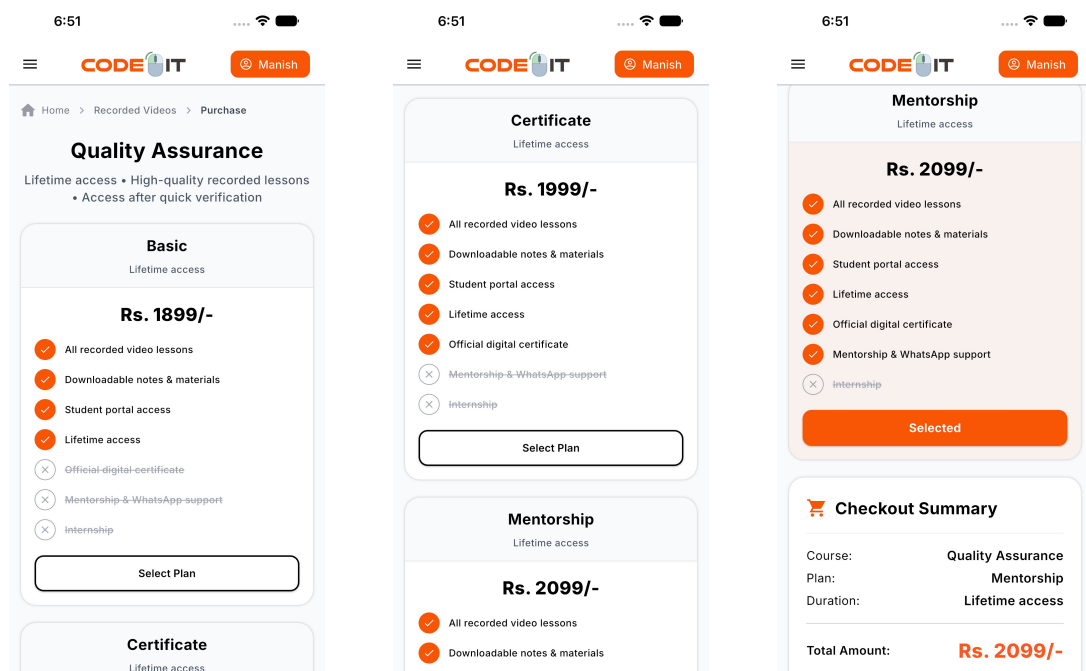


Figure J.1: Recorded Videos Purchase screen

Appendix K: Daily Work Log

Day	Date	Description of Task/Job done	Specific learned Knowledge and Skills	Approved by Supervisor
1	Apr 12, 2026	Attended online orientation meeting with supervisor. Discussed project timeline and goals.	Learned about company workflow and remote work rules.	
2	Apr 15, 2026	Joined official team chat groups. Created GitHub folder for code sharing.	Learned team communication tools and Git setup.	
3	Apr 16, 2026	Reviewed old web version of student portal. Listed features for mobile app.	Learned requirement analysis and feature mapping.	
4	Apr 17, 2026	Researched mobile design layouts. Wrote down initial project details.	Learned mobile UX planning basics.	
5	Apr 19, 2026	Planned screen navigation flow. Finalized folder structure for images and text.	Learned app architecture planning.	
6	Apr 20, 2026	Finished writing project layout document. Defined all app functions clearly about the app.	Learned technical documentation writing.	
7	Apr 21, 2026	Started designing app screens in Figma. Created user pathways for Dashboard.	Learned Figma prototyping and UX design.	
8	Apr 22, 2026	Made final design layouts for Dashboard and Home screen. Fixed spacing issues.	Learned UI mockup creation and design consistency.	
9	Apr 23, 2026	Checked all buttons and colors against requirements. Verified accessibility.	Learned design and color contrast checking.	
10	Apr 24, 2026	Sent final Project Plan and layouts to supervisor for approval.	Learned professional design handoff process.	
11	Apr 26, 2026	Created blank Flutter project template. Organized folders for MVCS structure.	Learned Flutter project setup and folder organization.	
12	Apr 27, 2026	Set up global fonts and color theme in Flutter. Configured base app settings.	Learned Flutter theming and app configuration.	

Day	Date	Description of Task/Job done	Specific learned Knowledge and Skills	Approved by Supervisor
13	Apr 28, 2026	Coded Support Page UI layout using Figma designs. Built basic widgets.	Learned Flutter widget implementation from designs.	
14	Apr 29, 2026	Cleaned up text boxes and category selectors. Added input validation.	Learned form validation and widget polishing.	
15	Apr 30, 2026	Added Support page to main navigation router. Tested screen transitions.	Learned Flutter navigation and route testing.	
16	May 1, 2026	Showed screens to team. Received new design instructions from office.	Learned how to update designs based on team feedback.	
17	May 3, 2026	Modified global app colors and buttons for new design changes.	Learned how to update app colors and buttons globally.	
18	May 4, 2026	Studied official company Figma files. Updated theme setups to match brand.	Learned design system interpretation.	
19	May 5, 2026	Rebuilt Support and Suggestion layouts completely. Removed old UI.	Learned design implementation and UI rebuilds.	
20	May 6, 2026	Finalized support forms with new company logos and icons.	Learned asset integration and brand alignment.	
21	May 7, 2026	Synced Support forms with top navigation menus. Tested structural links.	Learned navigation-integrated UI testing.	
22	May 8, 2026	Submitted rebuilt Support screens to manager for final check. Got approval.	Learned how to submit work for final approval.	
23	May 10, 2026	Programmed base HTTP networking rules using Dio package. Set up interceptors.	Learned REST API client setup and MVCS Pattern.	
24	May 11, 2026	Connected Support screen fields to server pipeline. Built SupportService class.	Learned how to connect app screens to server data.	

Day	Date	Description of Task/Job done	Specific learned Knowledge and Skills	Approved by Supervisor
25	May 12, 2026	Linked server data fields for Technical, General, and Counseling APIs.	Learned how to connect multiple API endpoints for different support types.	
26	May 13, 2026	Set up state management using Provider. Screens update automatically now.	Learned reactive state management patterns.	
27	May 14, 2026	Programmed OTP verification screen with timer. Added password reset logic.	Learned authentication flow and form validation.	
28	May 15, 2026	Created custom "View My Tickets" list view. Tested all APIs via Postman.	Learned ListView optimization and Postman testing.	
29	May 17, 2026	Sketched custom route navigation maps. Took over core navigation work.	Learned navigation architecture and task ownership.	
30	May 18, 2026	Continued navigation structural work. Documented route hierarchy for team.	Learned technical documentation during transitions.	
31	May 19, 2026	Coded slide-out hamburger menu and top app header bar.	Learned Drawer/AppBar implementation in Flutter.	
32	May 20, 2026	Created static layout for Terms of Service display. Added scrollable text.	Learned how to display fixed text like terms and service from the server.	
33	May 21, 2026	Merged navigation menus into main home screen. Fixed GitHub merge conflicts.	Learned Git conflict resolution and integration.	
34	May 22, 2026	Tested navigation tracks heavily. Fixed back-stack loop issues.	Learned navigation debugging and user flow testing.	
35	May 24, 2026	Listed dimensions of small and large screens. Found overflow errors.	Learned responsive design auditing methodology.	
36	May 25, 2026	Opened app on different test monitors. Documented formatting errors.	Learned cross-device compatibility testing.	

Day	Date	Description of Task/Job done	Specific learned Knowledge and Skills	Approved by Supervisor
37	May 26, 2026	Updated Support layout for dynamic scaling using MediaQuery.	Learned Flutter responsive layout techniques.	
38	May 27, 2026	Fixed long form fields to prevent red overflow flags on small phones.	Learned overflow prevention and adaptive sizing.	
39	May 28, 2026	Adjusted side menu spacing for tall aspect ratio devices.	Learned aspect-ratio-aware UI tuning.	
40	May 29, 2026	Confirmed all elements resize uniformly across aspect ratios. Final sign-off.	Learned comprehensive responsive validation.	
41	May 31, 2026	Sketched logic pathways for certificates. Developed "My Courses" grid layout.	Learned media handler planning and GridView layout.	
42	Jun 1, 2026	Verified server connection links for course modules. Tested data fetching.	Learned how to connect course screens to server data and test them.	
43	Jun 2, 2026	Set up code to email student completion certificates via backend trigger.	Learned email certificate integration.	
44	Jun 3, 2026	Prepared iOS 26 app icon and added in the flutter project.	Learned iOS asset catalog and icon generation.	
45	Jun 4, 2026	Fine-tuned iOS icons for light and dark display modes. Validated quality.	Learned platform-specific asset optimization.	
46	Jun 5, 2026	Met with team to check "My Courses" alignment. Fixed stray borders.	Learned collaborative UI review and refinement.	
47	Jun 7, 2026	Planned buffer targets for video players. Created clean video interface.	Learned video player architecture and streaming UI.	
48	Jun 8, 2026	Loaded active media viewer container with test streams. Validated playback.	Learned video player integration and testing.	

Day	Date	Description of Task/Job done	Specific learned Knowledge and Skills	Approved by Supervisor
49	Jun 9, 2026	Programmed Live Classes panel layout. Added active stream detection logic.	Learned live classes detection and dynamic highlighting.	
50	Jun 10, 2026	Built Notes Section for saving text-based class materials locally.	Learned local data persistence and note-taking UI.	
51	Jun 11, 2026	Cleared home screen network loading issues. Checked brand graphics.	Learned performance debugging and visual QA.	
52	Jun 12, 2026	Loaded app onto physical iPhone. Demonstrated stability to team lead.	Learned real-device deployment and demo skills.	
53	Jun 14, 2026	Researched device permissions for fingerprint/Face ID readers. Studied docs.	Learned native plugin integration and permissions.	
54	Jun 15, 2026	Met with supervisor. Received feedback to change 3D buttons to flat design.	Learned iterative design feedback alignment.	
55	Jun 16, 2026	Started rewriting styles across all files. Removed shadows and 3D borders.	Learned large-scale UI refactoring.	
56	Jun 17, 2026	Rewrote buttons and frames into clean flat aesthetic. Updated theme tokens.	Learned flat design and design token management.	
57	Jun 18, 2026	Integrated local_auth package into Flutter project.	Learned how to add backup login when biometric fails.	
58	Jun 19, 2026	Linked biometric sensor to login view securely. Tested on Android and iOS.	Learned native security integration and testing.	
59	Jun 21, 2026	Outlined testing setups for team. Drafted Course Purchase System data rules.	Outlined testing setups for team. Drafted Course Purchase System data rules.	
60	Jun 22, 2026	Continued drafting Course Purchase System UI rows and local state.	Learned e-commerce UI patterns and state drafting.	

Day	Date	Description of Task/Job done	Specific learned Knowledge and Skills	Approved by Supervisor
61	Jun 23, 2026	Helped QA team install Flutter tools on their computers.	Learned cross-functional technical support skills.	
62	Jun 24, 2026	Held training session for QA team. Explained features and page connections.	Learned how to explain app features to the QA team.	
63	Jun 25, 2026	Showed QA team how to use Postman for backend API testing.	Learned how to teach QA team to test APIs using Postman.	
64	Jun 26, 2026	Created and shared compiled APK file for QA real-device testing.	Learned release build generation and distribution.	
65	Jun 28, 2026	Formulated purchase listing flow. Coded buying layout for recorded courses.	Learned transaction flow implementation.	
66	Jun 29, 2026	Completed Recorded Courses Module buying layout with smooth UX.	Mastered UI and transactional design.	
67	Jun 30, 2026	Built dynamic search bar to filter course lists instantly. Used debouncing.	Learned real-time search filtering and optimization.	
68	Jul 1, 2026	Programmed attachment button for bank receipt uploads. Added compression.	Learned file upload and image compression handling.	
69	Jul 2, 2026	Placed receipt-upload tool into recorded videos view for easy access.	Learned component reusability and consistency.	
70	Jul 3, 2026	Cleaned up buy option parameters. Deleted repetitive layout code lines.	Learned how to clean up code and remove repeated lines.	
71	Jul 5, 2026	Grouped QA-reported bugs into checklist. Read official error sheets.	Learned how to sort and list bugs reported by QA team.	
72	Jul 6, 2026	Added "no internet connectivity" package for offline user alerts.	Learned global network monitoring and offline UX.	

Day	Date	Description of Task/Job done	Specific learned Knowledge and Skills	Approved by Supervisor
73	Jul 7, 2026	Fixed text rendering errors in course lists. Resolved sorting filter crash.	Learned how to fix list display errors and sorting crashes.	
74	Jul 8, 2026	Patched file system error causing duplicate downloads. Used unique naming.	Learned file conflict prevention and deduplication.	
75	Jul 9, 2026	Fixed authentication loop errors. Repaired password changing screen logic.	Learned auth state debugging and token management.	
76	Jul 10, 2026	Ran full regression tests. Submitted completed bug-free app to company.	Learned regression testing and production submission.	
77	Jul 12, 2026	Completed final handover documentation. Submitted all source code and assets to supervisor.	Learned how to properly finish and submit a project.	